VENDOR RECONCILE ? Current State Blueprint

Live SAP S21-Connected Vendor Reconciliation & Engine

LIVE INTEGRATION UPDATE

Version 2.0 ? September 2026

Executive Summary: The Vendor Reconcile platform has successfully achieved live data integration with the SAP S21 ERP system. The Executive Analytics Dashboard, dynamic 3-way matching engine, and vendor exception workflows are now powered by 100% real-time data, completely replacing legacy mock placeholders. It accurately tracks P2P lifecycle discrepancies natively via OData endpoints.

SECTION 1 ? LIVE SAP INTEGRATION CAPABILITIES

The Node.js (SAP CAP) backend securely interfaces with the SAP S21 System to automatically synchronize and process operational Procurement & AP data. We have implemented robust API handling for live Master Data retrieval, ensuring the Executive Dashboard reflects genuine transactional reality rather than simulated environments.

? True Material Overrides

Invoice line items accurately cross-reference active SAP Purchase Orders to fetch genuine material codes (e.g., 10314, PA30PALL) instead of static fallbacks.

? Real-Time Vendor Names

Dynamically queries the S21 Purchase Order API (A_PurchaseOrder) to extract true Supplier Address Names to instantly overwrite legacy DB numeric codes.

? Persistent Synchronization

Migrated backend to a highly durable local SQLite transactional database. All synced SAP records and user attachments now comfortably survive server restarts.

SECTION 2 ? EXECUTIVE DASHBOARD AUTOMATION

? Missing in File (Uninvoiced GRs)

Actively scans the live SAP **GrirSnapshot** table to mathematically highlight open Goods Receipts that have not yet been billed by suppliers.

? Missing in S/4HANA

The matching engine precisely flags imported vendor invoices where no corresponding SAP PO can be located (UNMATCHED/PO-NOT-FOUND).

? High-Value Exception Engine

Automatically filters and escalates systemic price or quantity mismatches exceeding the critical ?50,000 internal tolerance threshold.

? Dynamic Vendor Exceptions

Accurately sorts and maps specific real vendor names responsible for active discrepancies directly against the live matching engine rule violations.

SECTION 3 ? CURRENT TECHNOLOGY OVERVIEW

FRONTEND	BACKEND	SAP S21 INTEGRATION	DATA STORE
SAP Fiori Elements	**Node.js ? SAP CAP**	**OData / REST**	**SQLite / HANA-Ready**
HTML5, UI5 Responsive UX	Cloud Application Programming	Basic Auth, API Proxy	Full Transactional Durability

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SECTION 4 ? END-TO-END MATCHING ENGINE WORKFLOW

STEP 1
SAP PO & GR Sync
Extracts Live Data

?

STEP 2
Vendor Upload
Extracts Invoice

?

STEP 3
3-Way Verification
Applies Thresholds

?

STEP 4
Exception Flagging
Triggers Workflows

SECTION 5 ? SAP API MAPPING (S21)

Functional Area	SAP S21 Endpoints Integrated	Data Governance
Purchase Orders (ME23N)	`API_PURCHASEORDER_PROCESS_SRV/A_PurchaseOrder`	Source of truth for expected vendor, material, and agreed pricing.
Supplier Invoices (MIRO)	`API_SUPPLIERINVOICE_PROCESS_SRV/A_SupplierInvoice`	Source of truth for submitted billing amounts and vendor identification.
Goods Receipts (MIGO)	`API_MATERIAL_DOCUMENT_SRV/A_MaterialDocumentItem`	Source of truth for physical warehouse delivery quantities.

SECTION 6 ? AUTOMATED VERIFICATION OUTCOMES

? MATCHED / 3-WAY-CLEAR

Invoice successfully matched against Purchase Order and Goods Receipt within acceptable price/qty tolerances. Ready for payment clearing.

? UNMATCHED / PRICE-DIFF

Invoiced Amount drastically exceeds the expected Purchase Order rate beyond the configured 2% organizational tolerance.

?? UNMATCHED / QTY-EXCEEDS-GR

The vendor is billing for a higher quantity than what was physically received and logged by the GR specialist in the warehouse.

? UNMATCHED / PO-NOT-FOUND

The vendor invoice was successfully uploaded, but no active Purchase Order reference exists within the SAP S21 integration layer.